Sebastian Acendra

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EDUCATION

GEORGIA INSTITUTE OF TECHNOLOGY, Atlanta, GA — Graduation: December 2026
Bachelor of Science in Mechanical Engineering
Relevant Coursework: Machine Design, Heat Transfer, Thermodynamics, Fluid Mechanics, System Dynamics & Controls, Creative Decisions & Design, Mechanics of Deformable Bodies, Dynamics of Rigid Bodies, Differential Equations, Computing Techniques, Circuits & Electronics

GEORGIA STATE UNIVERSITY, Atlanta, GA — 2022–2023
Chemical Engineering Coursework — GPA: 4.00/4.00

ENGINEERING EXPERIENCE

DRONE DUCK HUNT, Georgia Tech — August 2026–Present
Mechanical Team Lead, Interdisciplinary Capstone Team

- Lead mechanical design for an Army-backed project in a 100+ student capstone: reusable hit-reporting hardware & software for aiding military personnel in training to shoot down drones.
- Owned module packaging, mounting concepts, and hardware fit for the electro-mechanical assembly, coordinating enclosure and mount geometry with the electrical team designing the sensor PCBs.
- Helped manage $2,047 against a $2,500 budget for two prototypes.
- Utilized Raspberry Pi 3 & camera to detect the orientation of an M4 rifle, as well as CAD and fabricate the weapon mount
- Field-tested the first prototype on-site in Oregon, and now leading bench testing to select the final design.

REFRIGERATION SYSTEM CHARACTERIZATION, Georgia Tech — April 2026–July 2026
3-Person Thermal Systems Team

- Proved an onboard compressor could replace building air supply & flagged bypass heat paths of a compressor to vortex tube system to ensure max amount of cold temperature possible.
- Identified supply pressure as the primary cooling lever: raising the pressure (∼33 psi) cut the fitted cooling time constant 42% and lowered projected steady-state load temperature by ∼8 K.
- Recommended an operating range from a five-point motor sweep (0.5 V - 2.5V), trading lowest cold-air temperature against 8.6% more cold-side airflow;

DRONE WING VIBRATION MITIGATION, Georgia Tech — January 2026–May 2026
3-Person Dynamics & Vibrations Team

- Diagnosed the root cause of wing-support failure by triangulating natural frequency through theory, impulse, white-noise, and sinusoidal testing
- Cut displacement-to-force gain 99.9% at critical frequency, providing anti-resonance by tuning a wing-extension mass absorber.
- Established a new operating range with at least 30 dB attenuation from the original resonance, keeping secondary resonances outside the wing's operating range.

GEORGIA TECH ECOCAR, Atlanta, GA — January 2024–May 2024
Mechanical Design / HV Box Team Member

- CAD-modeled custom HV-box mounts for milling around surrounding vehicle components, meeting packaging limits and electrical and mechanical clearances without interfering with adjacent systems.
- Verified mount load capacity with Altair HyperWorks FEA under vehicle G-force load cases so the brackets could withstand dynamic driving loads in the competition vehicle.

PROJECTS

THERMAL IMAGING DEVICE — August 2026–Present
Independent Thermal Systems Project

- Building a low-cost handheld thermal imager in a custom 3D-printed enclosure.
- Planning to use Python to create a heatmap with interpolation to aid with image smoothing.

PORTABLE LINUX CYBERDECK — September 2026–Present
Independent Mechatronics Project

- Designing a clamshell Linux-based terminal inside a Pelican hard case.
- CAD modeling electrical modules from measured components, connector exits, and cable-bend clearances to ensure all electronics fit the case's internal bounds on a $100 budget.

ADDITIONAL SKILLS

Technical: Mechanical design, CAD modeling, FEA setup, thermal analysis, vibration testing, electro-mechanical packaging, mechatronics, sensor integration, data acquisition, rapid prototyping
Software: Siemens NX, SolidWorks, Fusion 360, Altair HyperWorks, MATLAB, Python, Arduino IDE, Microsoft 365, Git
Programming: MATLAB data visualization, Python, Arduino/C++ control logic, basic OpenCV-based vision workflows
Languages: English (fluent), Spanish (fluent)